

Exercise 9

1. What is the difference between internal and external cluster validation indices?

Given are 15 instances $\{x_1, \dots, x_{15}\}$ and three classes $\{A, B, C\}$. The class assignment is as follows:

- A : $\{x_1, x_2, x_3, x_4, x_5\}$
- B : $\{x_6, x_7, x_8, x_9, x_{10}\}$
- C : $\{x_{11}, x_{12}, x_{13}, x_{14}, x_{15}\}$

Executing K-Means with (a) $K = 3$ and (b) $K = 5$ results in the two (imperfect) clusterings Z_1 and Z_2 :

- Z_1 :
 - $X_1 = \{x_1, x_2, x_3\}$
 - $X_2 = \{x_5, x_6, x_7, x_9, x_{10}, x_{15}\}$
 - $X_3 = \{x_4, x_8, x_{11}, x_{12}, x_{13}, x_{14}\}$
- Z_2 :
 - $X_1 = \{x_1, x_2, x_3\}$
 - $X_2 = \{x_5, x_6, x_7\}$
 - $X_3 = \{x_9, x_{10}, x_{15}\}$
 - $X_4 = \{x_8, x_{11}, x_{12}\}$
 - $X_5 = \{x_4, x_{13}, x_{14}\}$

Calculate the purity for each of the clusterings and compare the results. What is the problem of this measure?

2. Compute the Rand Index and Jaccard Coefficient of both clusterings in task 1.

Also, Calculate cohesion and separation for all clusters in Z_1 in task 1 assuming the data points have the coordinates as depicted in Table 1.

Use Euclidean distance as the distance measure. When calculating cohesion and separation, consider that the clustering was generated by K-means.

3. Assuming the dataset in Table 1 is given, calculate the silhouette coefficient for each of the three clusters and the whole clustering shown in Figure 1.

Table 1: Point coordinates for a 2-dimensional dataset.

ID	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	x_{11}	x_{12}	x_{13}	x_{14}	x_{15}
x	1	2	2	3	3	4	5	6	6	5	4	5	6	4	4
y	4	5	4	3	4	1	2	4	1	1	4	5	6	5	3

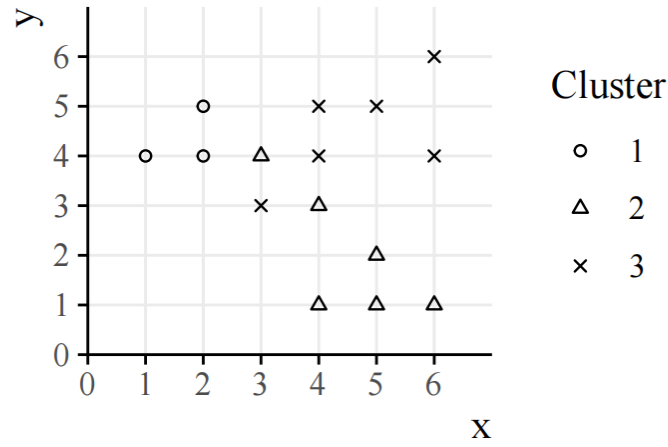


Figure 1: Clustering result on a two-dimensional dataset with 15 points.

4. What is Supervised learning, Semi-supervised learning, and Active Learning? How are they different from each other?
5. Briefly explain the following Active Learning Strategies
- Uncertainty Sampling
 - Query-by-Committee
 - Expected Model Change
 - Expected Error Reduction
 - Variance Reduction
 - Density-Weighted Methods